Lecture 00 / teaching notes

11 slides • 15-minute opening within the 6:30–10:00 pm joint session. Suggested slide delivery: 11:45; the remaining 3:15 allows student responses and the handoff.

Shared session plan (pm)

Time	Deck	Focus
6:30–6:45	Lecture 00	Nontechnical introduction
6:45–7:40	Lecture 01	Foundations, shift, substitution, frequency analysis
7:40–7:50	Break	10 minutes
7:50–8:45	Lecture 01	Vigenère, Kasiski, IC, key recovery
8:45–8:55	Break	10 minutes
8:55–9:30	Lecture 01	Playfair
9:30–9:50	Lecture 01	Transposition
9:50–10:00	Lecture 01	Comparison and questions

Suggested pacing within the opening

Slide	Main idea	Time	Elapsed
1	Cryptography	0:25	0:00–0:25
2	You used it before class.	0:55	0:25–1:20
3	Three questions behind digital trust.	1:30	1:20–2:50
4	New tools. New ways to communicate.	0:55	2:50–3:45
5	Private words. Public networks.	1:20	3:45–5:05
6	How do strangers trust a connection?	1:20	5:05–6:25
7	One tap. A chain of trust.	1:05	6:25–7:30
8	Your face can unlock a key.	1:10	7:30–8:40
9	Trust travels with the data.	1:05	8:40–9:45
10	Cryptography is part of the system.	1:25	9:45–11:10
11	Next: ciphers you can see inside.	0:35	11:10–11:45

Keep the cover brief. Invite one or two everyday examples on slide 2, avoid protocol tutorials, and use the final slide to hand over to Lecture 01’s formal vocabulary and cipher constructions.

Explanations and accuracy notes

The three protection goals

Confidentiality asks who can read the content. Integrity asks whether it has been altered. Authentication asks who or what is on the other side, or where a message came from. Cryptanalysis studies how protections can fail. These are different questions; secrecy alone does not answer all of them.

History and changing technology

Connect hand writing, machine-assisted military communication, and digital networks. Enigma cryptanalysis drew on Polish work and multinational Allied teams. Avoid attributing the history to a single person. The historical thread should motivate the subject, without adding a new cipher lesson.

Messaging and the web

The messaging illustration describes an end-to-end encrypted conversation, not all chat or ordinary email. The forwarding service is not intended to have the keys needed to read the content. Metadata, backups, and compromised devices are separate concerns. HTTPS authenticates the named domain and protects a connection; it does not certify the operator's honesty. The domain bank.example is fictional.

Payments and passkeys

Cryptographic payment checks help authenticate transaction data and detect changes. Payment methods do not all use identical mechanisms, and these checks do not prevent every fraud. With passkeys, a local face, fingerprint, or PIN check can authorize use of a credential; the website receives cryptographic proof, not the biometric as its login credential. Some passkeys sync between devices.

Passports and signed updates

A signature can support checks of data origin and integrity. An electronic passport's chip-data signature does not by itself identify the person holding the document or prove that a chip was not cloned. A software signature is not a guarantee that the program contains no bugs. These qualifications belong in spoken explanation when relevant.

One clear limitations discussion

The transfer scam is illustrative, not a news report: a person can be manipulated into authorizing a genuine transaction whose cryptographic checks succeed. Secure devices, software, key handling, people, and processes also matter. Keep this point in one slide rather than repeating warnings after every application.

Bridge to the first technical lecture

The final glimpse MEET → PHHW uses a shift of three, but do not derive it here. Invite curiosity about hiding and recovering a message. Lecture 01 now begins with constructing and analysing ciphers; its repeated motivation is removed. It retains the precise vocabulary, shared-secret assumption, mathematical model, and worked examples.

Sources and updates from the reference

The source records distinguish material inspected in the uploaded reference, documentation retrieved from official repositories, and background standards cited bibliographically. The deck does not depend on live news or rapidly changing statistics.

[R] Teaching reference

Wu Hongjun, supplied Lecture01_Introduction.pdf, 12 pages.

Uploaded reference inspected. Gist retained; original PDF not redistributed.

[1] HTTPS and TLS

Mozilla / MDN, HTTPS glossary and Transport Layer Security guide.

https://developer.mozilla.org/en-US/docs/Web/Security/Defenses/Transport_Layer_Security

Supporting text retrieved from the official mdn/content GitHub repository.

[2] Encryption and messaging

Mozilla / MDN, Encryption glossary; Signal Android official project README.

https://developer.mozilla.org/en-US/docs/Glossary/Encryption

Encryption definition and official messenger project description retrieved. The slide uses a generic end-to-end encrypted conversation, not a claim about all apps.

[3] Passkeys

W3C, Web Authentication specification; Mozilla / MDN, Passkeys guide.

https://developer.mozilla.org/en-US/docs/Web/Security/Authentication/Passkeys

W3C specification source and MDN guide retrieved from their official GitHub repositories.

[4] Signed updates

The Update Framework, specification, motivation and metadata verification.

https://theupdateframework.github.io/specification/latest/

Specification source retrieved from the official theupdateframework/specification GitHub repository.

[5] Payment and passport background

EMVCo, EMV chip/contactless specifications; ICAO Doc 9303, Part 12, Security Mechanisms for MRTDs.

Background standards cited bibliographically; no usage statistics, regional rollout dates, or claim of a newly published version.

What was refreshed

Retained the three security goals, the connection to computing and communication, military history, finance, daily-life applications, and the system-level limitation. Replaced the Swift-volume and regional ATM discussion with timeless payment concepts; refreshed account authentication with passkeys; added signed software updates; and used a clearly illustrative scam instead of relying on the 2016 Bangladesh Bank incident. The original uploaded PDF is not redistributed.